



Bridge of Spaghetti

PLANRS ACADEMY

Famous Indian Bridges

India's amazing bridges and structures!
Can you spot the shapes that make
them strong?



What You Will Learn Today

- Shapes like triangles make bridges strong
- What is compression (pushing) and tension (pulling)
- How to build a bridge with spaghetti and marshmallows
- Why a strong foundation matters



The King of Shapes: Triangle!

Triangles don't wobble – they stay strong! When you push on a triangle, it keeps its shape. But a square? It bends and falls over easily. That's why builders use triangles to make bridges super strong!



Compression and Tension

Compression = Pushing force (like when you press your hands together)
Tension = Pulling force (like when you pull a rope)
These forces help bridges stay strong!





Stability: Iron Pillar of Delhi

This pillar has stood strong for hundreds of years without bending! It shows us that a strong foundation keeps things steady and tall.

Arches vs. Straight Beams

Look at Howrah Bridge's strong arch shape compared to simple straight stone culverts. Arches help spread weight evenly so bridges don't break!



Triangles in Construction

Look at these amazing structures! The Pamban Bridge, mobile towers, and railway bridges all use triangles to stay strong. Triangles help these structures stay steady during wind and heavy trains!



Materials You Need

- Thick spaghetti strands (20-30 pieces)
- Marshmallows or clay balls (10-12 pieces)
- Small coins or erasers for testing weight
- Safety reminder: Only use these safe materials — no sharp tools!



Building Your Spaghetti Bridge

1. Connect spaghetti with marshmallows to make triangles
2. Build a bridge shape with a strong base
3. Use triangles to make it steady
4. Test your bridge by gently adding coins or erasers





Stress Test Time!

Let's see how many coins your bridge can hold without falling!
Think about monsoon winds shaking real bridges. Is your bridge strong enough?

Success Means...

- Spot a triangle in your bridge or a photo
- Build a bridge that holds at least 3 erasers or coins
- Explain what compression (pushing) means in your own words





Recap and What We Learned

Triangles make bridges strong and steady!
Compression means pushing, tension means
pulling.

Foundations keep bridges from falling down.
Indian bridges like Howrah use these smart ideas
every day!

Keep Building!

Thank You!

Great job today! Keep
exploring and building strong
bridges!

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